Experiment:13 Simulation-Based Frequency-Domain Characterization of a Microwave Two-Port Network Using S-Parameters and Network-Analyzer Emulation

Aim: To simulate a microwave two-port network using MATLAB and analyze its S-parameters and frequency response.

Apparatus:

Computer/Laptop
MATLAB software

Procedure:

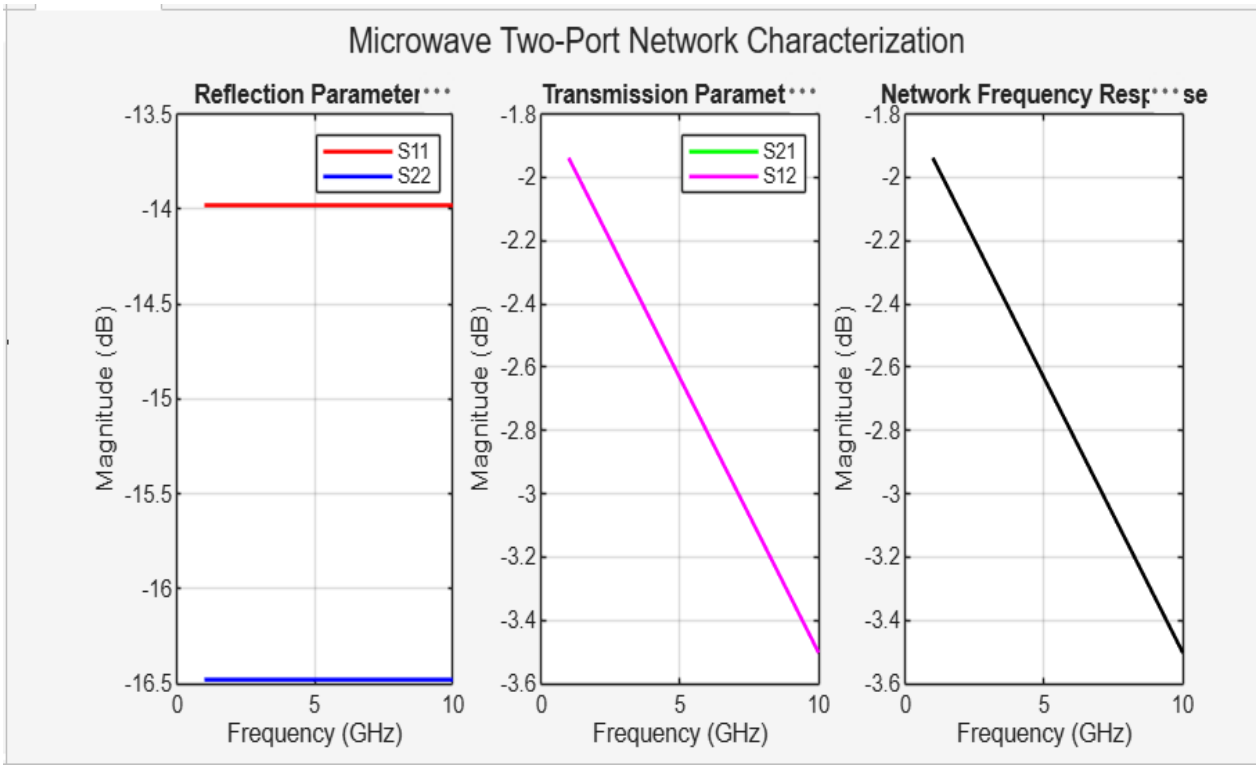
- Open MATLAB and create a new script.
- Define the frequency range and S-parameter values.
- Calculate the reflection and transmission coefficients.
- Plot S11, S21, S12, and S22 against frequency.
- Compare the input and output characteristics of the two-port network.
- Run the program and observe the results.

Code:

```
clc;
clear;
close all;
f = linspace(1,10,500);
S11 = 0.2*ones(size(f));
S21 = 0.8*exp(-0.02*(f-1));
S12 = S21;
S22 = 0.15*ones(size(f));
figure;
subplot(1,3,1);
plot(f,20*log10(S11),'r','LineWidth',1.5);
hold on;
```

```
plot(f,20*log10(S22),'b','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Reflection Parameters');
legend('S11','S22');
subplot(1,3,2);
plot(f,20*log10(S21),'g','LineWidth',1.5);
hold on;
plot(f,20*log10(S12),'m','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Transmission Parameters');
legend('S21','S12');
subplot(1,3,3);
plot(f,20*log10(S21),'k','LineWidth',1.5);
grid on;
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
title('Network Frequency Response');
sgtitle('Microwave Two-Port Network Characterization');
fprintf('Frequency Range = 1 to 10 GHz\n');
fprintf('S11 = %.2f dB\n',20*log10(0.2));
fprintf('S21 = %.2f dB\n',20*log10(0.8));
```

Output:



Result:

The microwave two-port network was simulated using MATLAB, and its reflection, transmission, and frequency-response characteristics were analyzed successfully.